

Influence of grading on the mechanical behaviour of Stava tailings

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Tailings dams can be fragile structures and too often they are subject to liquefaction. The material they are composed of is rather peculiar, having a man-made origin, and their behaviour is still not very well understood. The soil from the Stava tailings dams, structures that were subject to liquefaction in 1985 causing extensive destruction, was investigated in this study as an example of tailings. Two main aspects of their behaviour have been examined: the influence of the percentage of silt and sand that compose the soil on its mechanical behaviour, and the susceptibility to liquefaction, analysing the behaviour within a critical state framework. In this paper it is shown that, as the quantity of silt increases, the slope and intercept of the normal compression line and critical state line at higher pressures decrease until an inversion of behaviour is observed, while at lower stress levels the critical state line changes its position but not its slope. This means that the effect of adding silt on the critical state line location at low stresses is disconnected from that at high stresses. The analysis of liquefaction within a critical state framework was found to provide a simple explanation for the patterns of behaviour that are typical of static liquefaction. Three classes of behaviour related to the current void ratio and stress state of the soil were identified.

KEYWORDS: compressibility; dams; laboratory tests; liquefaction; sands; silts

INTRODUCTION

The Prestavel tailings dams were constructed in the Stava valley during the 1960s to the 1980s. The two dams and lagoons had a typical construction with the coarser sand fraction separated to create the embankment and then the finer fraction deposited behind the dam on a beach and pond. The more recent upper dam had been constructed further upslope with its toe on the lower pond, giving a total height of the two dams of 50 m and weakening even more a structure that was already unstable. The causes of the failure that occurred on 19 July 1985 are complex and have been investigated in detail by others (e.g. Chandler & Tosatti, 1995; Colombo & Colleselli, 2003), but the result was that the upper dam failed, promoting static liquefaction and causing a mudflow of 200 000 m³ that travelled down the valley at speeds of up to 90 km/h, killing 268 people. Although the instability of the structure's design played the main role in the accident, the nature of the tailings itself makes this material particularly subject to liquefaction pro-

Les barrages de stériles miniers sont parfois des structures fragiles, trop souvent sujettes à la liquéfaction. Plutôt insolites, ils sont composés de matières produites par l'homme dont le comportement est toujours mal connu. La présente communication se penche, à titre d'exemple de stériles miniers, sur le sol du barrage de stériles miniers de Stava, une structure qui subit une liquéfaction en 1985 en provoquant des dégâts considérables. Les deux aspects principaux de ce comportement ont été examinés, à savoir : influence du pourcentage de boue et de sable dans la composition du sol sur son comportement mécanique, et sa susceptibilité à la liquéfaction, en procédant à l'analyse du comportement dans le cadre de l'état critique. Dans le présent article, on démontre qu'au fur et à mesure de l'augmentation des boues, la pente et l'intersection de la ligne de compression normale et de la ligne d'état critique aux pressions élevées diminuent jusqu'à ce que l'on relève une inversion du comportement, tandis qu'à des niveaux de contrainte inférieurs, la ligne d'état critique change de position, mais non pas son inclinaison. Ceci signifie que l'adjonction de boue sur l'emplacement de la ligne d'état critique en présence de faibles contraintes n'est pas en rapport avec l'effet en présence de contraintes élevées. On a relevé que l'analyse de la liquéfaction dans un cadre d'état critique fournit une simple explication des formes de comportement typiques de la liquéfaction statique. On a identifié trois classes de comportement relatif à l'indice de vide actuel et de l'état des contraintes du sol.

blems. Accidents resulting from the liquefaction of tailings are, unfortunately, still very common. Among the most disastrous examples of tailing dam failure are Mufulira, Zambia, in 1970, and Marriespruit, South Africa, in 1994 (Blight & Fourie, 2003).

A particular feature of tailings dams is that the grading of the material varies with location, not only between the dam and the lagoon, but also within the lagoon with increasing distance from the point of the dam where the slurry of fines and water is pumped into the lagoon. Typically both the coarse and fine fractions tend to be non-plastic since they are the products of crushing and processing of a parent rock. The research presented in this paper is a general study of the effects of grading on the mechanics of non-plastic soils, with an emphasis on understanding the principal causes of static liquefaction in soils of varying grading, rather than a specific analysis of the particular causes of the Stava disaster.

Despite being the subject of intensive research, the factors causing static liquefaction remain elusive and complex. There is general agreement that soils with their in situ states above the critical state line (CSL, or steady-state line) in the $e: \ln(p')$ plane, with positive state parameters, are more prone to liquefaction, as illustrated in Fig. 1 (e.g. Yang, 2002). Critical state lines for sands are generally only linear in the $e: \ln(p')$ plane at high stress levels, and are typically curved lines that tend to a horizontal asymptote as p' reduces (e.g. Verdugo & Ishihara, 1996), often modelled as

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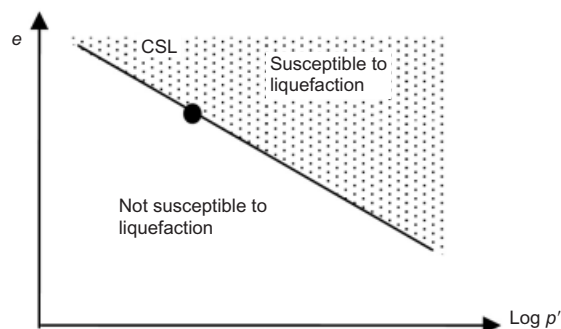


Fig. 1. Criterion for flow liquefaction susceptibility based on state parameter concept (based on Kramer, 1996; Been & Jefferies, 1985; and others)

bilinear (e.g. Been *et al.*, 1991). Although a state parameter has been used for liquefaction, there is less discussion in the literature about whether this should be applied to a curved CSL and, if so, whether the state parameter is measured with respect to the straight CSL at high pressures or to the flatter part at low pressures. There is also some ambiguity about the definition of liquefaction. For some authors it simply indicates strain softening and compressive undrained behaviour, so that a sample under stress control would fail uncontrollably at the peak deviator stress, $q_{\max}$ (e.g. Sladen *et al.*, 1985). In other cases authors have defined it to indicate that the soil has genuinely reached a liquefied state, with zero effective stress and zero strength (e.g. Yamamuro & Covert, 2001).

The evolution of the shape and location of CSLs in the $e: \ln(p')$ plane as fines are added to change the grading of a sand has received increasing attention in recent years. Again there is a far from clear picture, and often the findings of different authors seem contradictory, some reporting increases in the gradient of the CSL and others decreases. These apparent contradictions may simply reflect the different stress levels that have been investigated by different authors; while at higher stress levels the steeper part of the CSL, that may be dominated by particle breakage, has been shown to become flatter as fines are added (e.g. Coop & Atkinson, 1993), the flatter part of the line at lower stresses, where particle breakage is less significant, tends to become steeper (e.g. Been & Jefferies, 1985). Thevanayagam *et al.* (2002) found that as fines were added, the location of the CSL also initially moved downwards in the $e: \ln(p')$ plane, although the shape for different gradings remained similar, but then reversed direction to move upwards again. The grading that gave the lowest location was defined as the 'threshold fines content'.

There is also conflicting evidence as to whether the location of the CSL, and hence the susceptibility to liquefaction, is affected by the initial fabric created by the sample preparation method, with a unique CSL being determined by some (e.g. Jefferies & Been, 2006), while others (e.g. Chu *et al.*, 2003; Vaid & Sivathayalan, 2007) have found significant effects of fabric on liquefaction potential. An added complication is that there is increasing evidence that many gap-graded and intermediate graded 'transitional' soils have critical state and normal compression lines (NCLs) whose positions depend on the initial density and therefore probably on the fabric at deposition (e.g. Nocilla *et al.*, 2006; Ferreira & Bica, 2006).

The laboratory tests discussed in this paper were designed to investigate in more detail some of these issues. Soils from Stava were split into their two predominant gradings of sand and silt and recombined in different proportions to investigate the effect of grading on the behaviour, and in particular

their liquefaction potential. Oedometer tests were carried out to study the effect of grading on the compressibility of the soils and to see whether the soils were transitional or not, since a transitional behaviour adds another degree of complexity to the framework of liquefaction, given that in this case there is not a unique CSL. It will be shown that the soils are not transitional for any grading; therefore triaxial tests have been used to identify the CSLs for each grading and to examine liquefaction potential relative to the location of these lines.

MATERIAL PROPERTIES AND TESTING PROCEDURES

The Stava tailings used in this study were collected in 2005 from the lower portion of the upper dam, which remained in place after the collapse. They are composed essentially of two grain sizes: a fine sand and a silt. The two grain size distributions are shown in Fig. 2. Given the small quantity of clay present in the fine material, from now on the term 'silt' will be used as representative of the fines. Within the dam, the sandy material formed the embankment and the grading changed gradually towards a silty material inside the basin. All of the materials in the structure were very loose because of the method of deposition used; the sands were separated by hydro-cyclone before being deposited in the dam, while the silts were sedimented within a lagoon behind the dam.

The specific gravity G_s of the sand is 2.721 and that of the silt is 2.828. The liquid and plastic limits of the silts are $w_L = 27.4\%$, $w_P = 18\%$, giving a plasticity index $PI = 9.4\%$. An X-ray diffraction (XRD) analysis showed that the mineralogy is almost identical for the two gradings. Both were predominantly quartz, with 81% for the sand and 75% for the silt, with about 9–10% calcite in each, and smaller amounts of illite/mica, K-feldspar and kaolinite. The only significant difference was in the amount of fluorite (11% in the silt and 3% in the sand). The nature of the particles was also analysed by means of a QicPic, an image analysis apparatus that determines size and shape through laser scanning of the particles in two dimensions. The results indicated very similar shapes for the sand and silt with a mean sphericity of 0.88, where sphericity is the perimeter of a circular particle of the same area as the particle, divided by its actual perimeter. Both sizes had a mean aspect ratio of about 0.7. The only slight difference was in the convexity, which is the area of the particle divided by its area if any concavities within its perimeter are filled. This was 0.90 for the sand and 0.87 for the silt. The data therefore indicate that both the sand and silt have particles of sub-angular or sub-rounded shapes, tending to be only slightly elongated. The characteristics of this material are highly influenced by

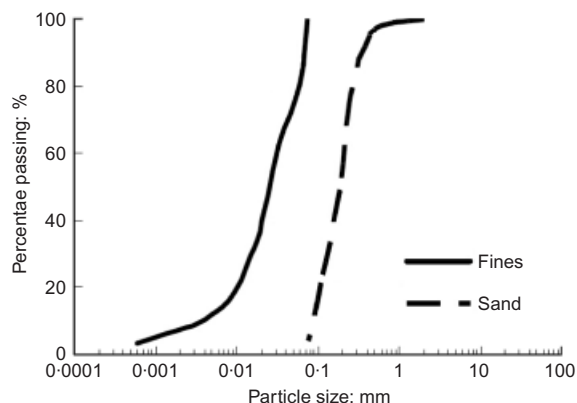


Fig. 2. Grain size distributions

its man-made origin, and the fact that the finer material has not been subject to chemical weathering explains its low plasticity and the similarity of its mineralogy to that of the sand.

The maximum and minimum void ratios of the sand were measured using ASTM D4253 (ASTM, 2002a) and D4254 (ASTM, 2002b) procedures. They were, respectively, 1.068 and 0.615. As the ASTM standards are not suitable for sands containing more than 15% fines, the void ratio extremes of other mixtures of sand and silt could not be measured following these standards.

A series of oedometer and triaxial tests was carried out on the sand and silt mixed at different percentages. The oedometer tests were carried out in oedometers with diameters of 38 mm and 50 mm with a maximum applied load of approximately 14 MPa; most of the samples were prepared using the moist tamping technique. This method allows very loose samples to be created, and thus liquefaction to be investigated. In addition, the structure created is likely to be similar to that of the moist, unsaturated materials that were placed in the Stava embankment. The mixtures investigated were: pure sand, 90% sand–10% silt, 70% sand–30% silt, 50% sand–50% silt, 30% sand–70% silt and pure silt. The soil mixtures with a silty fines content between 0 and 30% are representative of the material that formed the dam structure; the pure silt corresponds to the material that formed the ‘pond’; the mixtures with 30% to 70% silts are representative of the zone of the dam called the ‘beach’, the higher the silt content, the closer to the pond. To check if there was any transitional behaviour with non-unique NCLs, for each mixture a range of densities was tested, trying to achieve also the loosest and the densest possible initial states.

The triaxial cells used were all stress path controlled, and were equipped with internal axial and sometimes radial displacement transducers. The sample dimensions were either 38 mm in diameter and 76 mm high or 70 mm in diameter and 140 mm high. The mixtures investigated through triaxial testing were the clean sand, 70% sand–30% silt (indicated as 7030), 50% sand–50% silt (5050) and the pure silt. Again, almost all the samples were prepared using the moist tamping technique. However, some samples were prepared with different techniques such as wet and dry pluviation to study whether this would affect the position of the CSL in $e: \ln(p')$ space. Each sample was saturated under back pressure until a B value of 0.96 was achieved. Most of the specimens were initially compressed isotropically and then sheared under undrained conditions. In the data analysis, there was no need to correct for the membrane penetration effect because of the small particle sizes, but the membrane rigidity strongly affected the results at low mean effective pressures. To correct the stresses, the formulae suggested by Fukushima & Tatsuoka (1984) were used

$$\begin{aligned} \Delta\sigma_{ax} &= 0 \\ \Delta\sigma_{rad} &= -2 \left[\frac{M \varepsilon_{rad}}{D_i} \right] \end{aligned} \quad (1)$$

where σ_{ax} and σ_{rad} are the axial and radial total stresses, D_i is the initial diameter of the sample and M is the extension modulus of the rubber membrane per unit width ($= 0.36 \text{ kN/m}$). It should be borne in mind, however, that at extremely low effective stresses, below 10 kPa, the accuracy of all stress measurements will be poor as the uncertainty in the correction is too large. A list of the triaxial tests performed and details about the testing procedures are given in Table 1.

ONE-DIMENSIONAL COMPRESSION

The oedometer tests performed on different mixtures of sand and silt show that at higher stress levels it is possible to identify a unique one-dimensional normal compression line (1D-NCL) for each mixture, as can be seen in Fig. 3. This means that Stava tailings are not examples of transitional soils, despite the gradings of the material being very similar to those of other soils that have shown transitional behaviour (e.g. Nocilla *et al.*, 2006). The reasons for this are not clear yet and need to be further investigated. However, there is a larger scatter of the curves from tests on 50% sand–50% silt, giving a less clear convergence to a unique 1D-NCL, which might indicate a slight tendency towards transitional behaviour.

Comparing all the 1D-NCLs from the different sand–silt mixtures, it can be seen (Fig. 4(a)) that the 1D-NCL moves downwards as the silt content increases, and the slope, C_c , decreases (Fig. 4(b)), but this tendency inverts when a silt content between 50% and 70% is reached. The reversal of trend is also seen in the intercept of the NCLs at 1 kPa and the average of the initial void ratios of the mixtures that could be made. It should be noted that, since the mineralogy and the particle shapes of the sand and silt fractions are very similar, the position and gradient of the NCLs cannot depend on differences in the nature of the particles, but solely on the grading of the mixture.

The silty fines content corresponding to the inversion of behaviour will be referred to as the ‘transitional fines content’ (TFC) as previously used by Yang (2004). Inversions of trends in mechanical behaviour at a certain fines content have been previously observed in the literature by, among others, Pitman *et al.* (1994), who found a maximum decrease in the void ratio under isotropic compression at a fines content of 20%, and Thevanayagam *et al.* (2002), who observed an inversion of the trend of the CSL for a fines content of 40%. Polito (1999) recognised a similar feature and called it the ‘limiting fines content’. It must be noted, however, that for the Stava tailings the TFC is much higher than is commonly found in other soils, that is between 25% and 45% (Polito & Martin, 2001).

Following the idea that the usual void ratio is not able to describe the behaviour of a sand containing fines, many authors (e.g. Thevanayagam *et al.*, 2002; Ni *et al.*, 2004) have successfully identified unique behaviour, regardless of the fines content, through the use of the ‘equivalent intergranular void ratio’, e_{sk-eq} . This parameter is a modified form of skeletal or granular void ratio (Mitchell, 1976) that accounts for the different effects of plastic and non-plastic fines. It is defined as (e.g. Ni *et al.* 2004)

$$e_{sk-eq} = \frac{e + (1 - b)fc}{1 - (1 - b)fc} \quad (2)$$

where $1 - b$ is the fraction of fines that can be considered as potential void space, and $b \leq 1$, with negative values of b when the fines are plastic. The TFC is then a threshold above which the e_{sk-eq} cannot be used successfully, as the matrix is formed by fines. The e_{sk-eq} parameter was used to re-analyse the data in Fig. 4 in order to see whether the compression behaviour of the Stava silty sands could be normalised in this way and might then converge with the compression curve of the clean sand. Only the mixtures with a fines content $\leq$ TFC, that is 50%, were considered. A value of $b = 0.8$ was found to be optimum, as the curves all then converged to a common point. However, as can be seen in Fig. 5, even with the optimum value of b , the 1D-NCLs were found to be far from converging to a unique line and thus creating a unique relationship between e_{sk-eq} and p' , but they rather tend to cross each other. It is possible that the

Table 1. List of the triaxial tests

No.	Type of test	Material used	p'_c : kPa	q_c : kPa	e_c	p'_{cs} : kPa	q_{cs} : kPa	e_{cs}	Method
001	CID	Clean sand	100	0	0.911	187.1	262.9	0.896	MT
002	CID	Clean sand	300	0	0.889	571	813	0.820	MT
003	CIU	Clean sand	100	0	0.933	95.3	135.6	0.933	MT
004	CIU	Clean sand	300	0	0.915	195	280.2	0.915	MT
005	CAU	Clean sand	100	100	0.943	101.2	144.1	0.943	MT
006	CIU	Clean sand	600	0	0.872	308	448.3	0.872	MT
007	CID, Vconst	Clean sand	300	0	0.914	202.9	293.7	0.914	MT
013	CIU	Clean sand	300	0	0.939	73.8	116.1	0.939	MT
014	CIU	Clean sand	300	0	0.751	1045	1475	0.751	WP
015	CIU	Clean sand	300	0	0.958	62.8	91.5	0.958	MT
017	CIU	70% sand, 30% silt	300	0	0.733	L	L	0.733	MT
019	CIU	70% sand, 30% silt	300	0	0.651	65.3	98.4	0.651	MT
020	CIU	Clean sand	300	0	0.716	1078.8	1563.4	0.716	MT
021	CIU	70% sand, 30% silt	300	0	0.544	517	750	0.544	WP
022	CIU	Silt	300	0	0.726	99.9	138.3	0.725	MT
023	CIU	Clean sand	100	0	0.759	900	1320	0.759	WP
024	CIU	50% sand, 50% silt	300	0	0.600	35	50	0.600	MT
025	CIU	50% sand, 50% silt	100	0	0.601	23	33	0.601	MT
026	CIU	Silt	100	0	0.695	123	167.7	0.695	MT
027	CIU	Clean sand	100	0	1.022	L	L	1.022	MT
028	CIU	50% sand, 50% silt	150	0	0.663	12.7	16.8	0.663	MT
029	CIU	70% sand, 30% silt	75	0	0.748	L	L	0.748	MT
030	CIU	Silt	100	0	0.775	15.8	18.6	0.775	MT
032	CIU	70% sand, 30% silt	150	0	0.745	L	L	0.745	MT
033	CID	Silt	400	0	0.719	720.3	959.5	0.657	MT
034	CID	50% sand, 50% silt	400	0	0.593	770	1111.7	0.535	MT
036	CAU	70% sand, 30% silt	254	95	0.727	L	L	0.727	MT
037	CIU	Clean sand	75	0	0.839	465	645	0.839	DP
038	CIU	Clean sand	75	0	0.993	7.8	9.5	0.993	MP 1% wc
039	CIU	Silt	75	0	0.832	L	L	0.832	MT
040	CIU	50% sand, 50% silt	75	0	0.658	L	L	0.658	MT
041	CAU	70% sand, 30% silt	223	144	0.666	44.3	65.6	0.665	MT
043	CIU	70% sand, 30% silt	75	0	0.644	153	230	0.644	DP
044	CID	70% sand, 30% silt	700	0	0.579	1350	1957.5	0.459	MT
045	CID	Silt	650	0	0.675	1220	1708	0.614	MT

Note: Test 007 was a drained, constant volume test. e_c , p'_c and q_c are void ratio, mean effective stress and deviator stress at the end of the consolidation stage. e_{cs} , p'_{cs} and q_{cs} are the same parameters at the critical state. L: test that reached liquefaction; MT: moist tamping; WP: wet pluviation; DP: dry pluviation; MP 1% wc: dry pluviation, but less than 1% in weight of water was added to create a looser sample.

lines might be made to converge if b , and therefore the influence of the different gradings, was assumed to vary with the stress level, but this has not yet been addressed in the literature and so the choice of the variable b would simply be guesswork. The value of b found is also much greater than the recommendations from Rahman & Lo (2008), who, on the basis of an extensive literature survey, suggest a value of b between 0.08 and 0.43 for non-plastic fines. Rahman & Lo (2008) also suggested that b depends on the grain size of the fines relative to that of the host sand, for which they give a relationship; however, even using this method, it was still not found possible to produce a unique compression curve. From this it can be concluded that while b allows for a qualitative estimation to be made of to what extent the fines and the sand skeleton support the stress, it actually has no physical meaning, and the behaviour of different mixtures cannot always be normalised to a unique behaviour simply through the use of e_{sk-eq} .

IDENTIFICATION OF THE CRITICAL STATE LINES

To investigate the critical states of the different mixtures, triaxial tests were performed trying to achieve the maximum axial strain in each test that was possible in the apparatus, generally around 20–30%. Unfortunately, a critical state was not always reached and it was necessary sometimes to extrapolate the data to a ‘most probable’ value. Fig. 6 shows

an example of a test that was stopped when the stress ratio and the pore pressure variation were still changing slightly. The value of stress ratio at the critical state, $(q/p')_{cs}$, has been predicted from the stress–dilatancy graph for the drained tests, extending the last part of the curve to the point of intersection with the zero dilatancy axis. For undrained tests, the rate of change of excess pore pressure, $\delta(\Delta u)/\delta\epsilon_s$, plotted against the stress ratio, was used to assess dilatancy, where Δu is the change in pore pressure and ϵ_s is the shear strain ($=\epsilon_a$ for undrained conditions), as in the example in Fig. 6(b). The value of p'_{cs} was then evaluated from the q – p' graph, once $(q/p')_{cs}$ was known. Another problem that affected some tests was the localisation of the strains. For tests where this was observed, it has been attempted to identify the critical state from the data prior to localisation. An example is given in Fig. 7: the anisotropically consolidated test 005 reaches values of stress ratio and pore pressure that are nearly stable (Fig. 7(a)), while, in the rate of pore pressure change plot (Fig. 7(b)), the curve reaches zero rate of change. However, after a certain strain level, a localisation seems to appear, as q/p' starts to decrease without changing the rate of pore pressure change, and the pore pressure starts to increase indefinitely. In this case, as the sample seems to have reached a stable state before localisation, that state has been considered to be representative of the critical state. The results from such tests are generally in good agreement with those from other tests.

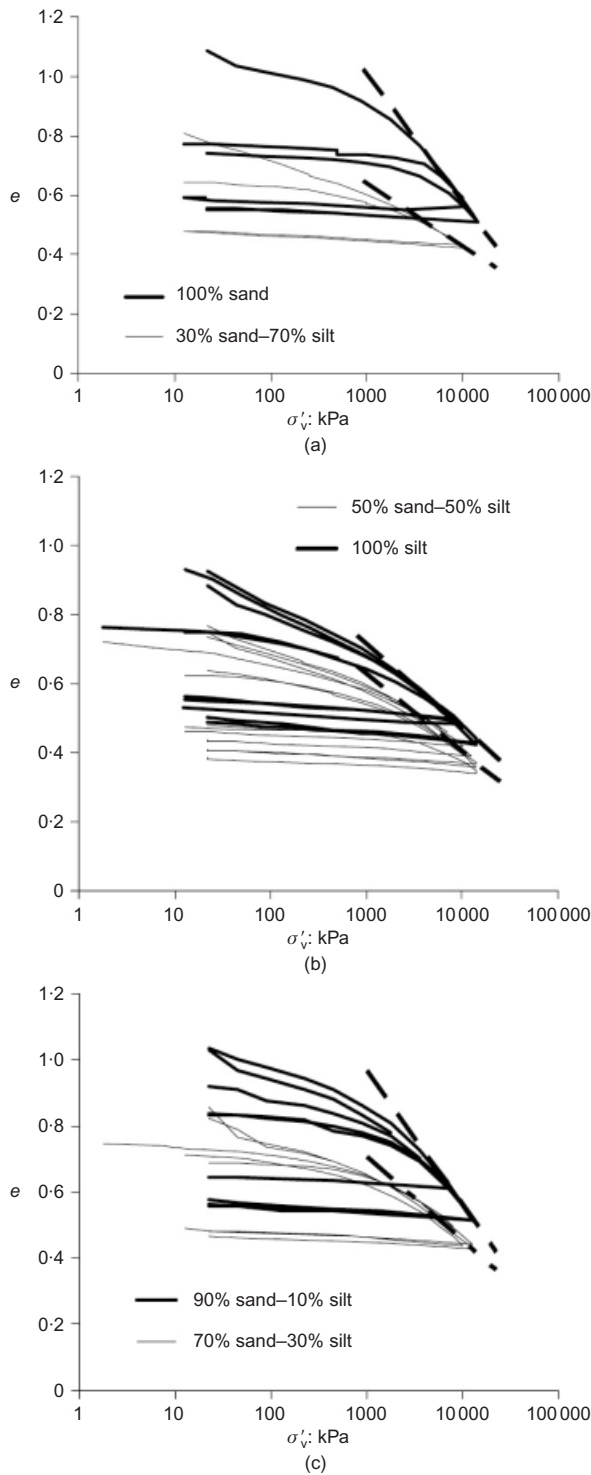


Fig. 3. Oedometer test results. Dotted lines identify the 1D-NCLs: (a) tests on clean sand and on 30% sand–70% silt samples; (b) 50% sand–50% silt and pure silt samples; (c) 90% sand–10% silt and 70% sand–30% silt samples

The critical state angle of shearing resistance, ϕ'_{CS} , was found to be about 35° for the sand and the 5050 material, 36° for the 7030 material and 34° for the silt, so that the value of ϕ'_{CS} varies only slightly with the mixture, as can be seen in Fig. 8. This is probably due to the common origin of the sand and the silt.

The CSLs of the different mixtures are shown in the e : $\ln(p')$ plane in Fig. 9, along with the 1D-NCLs obtained from the oedometer tests. The values of p' were estimated by assuming a K_0 equal to $1 - \sin(\phi'_{CS})$; any error in this

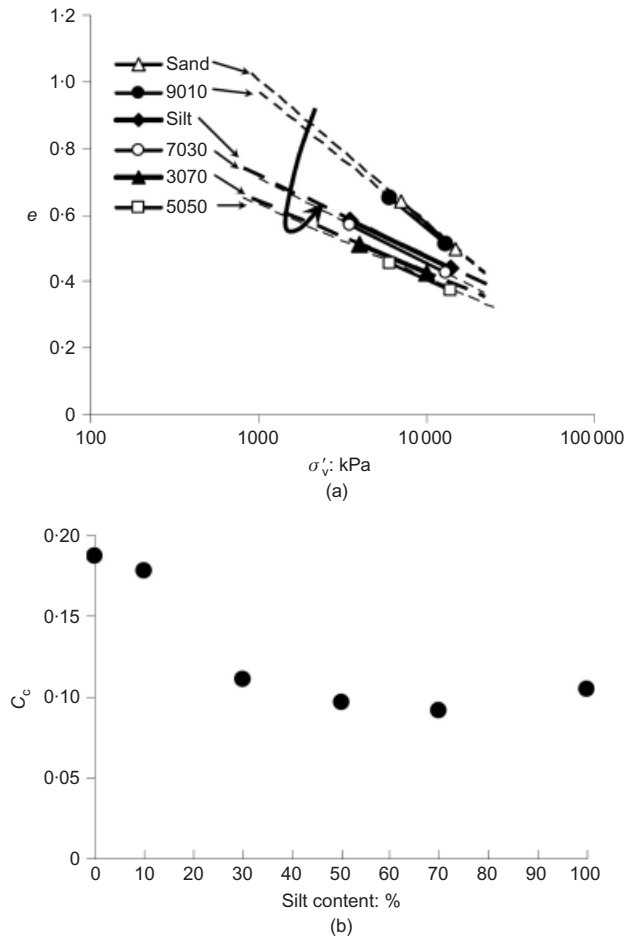


Fig. 4. (a) 1D-NCLs of the investigated mixtures of silt and sand (9010, sand with 10% silt; 7030, 70% sand and 30% silt; 3070, 30% sand and 70% silt; 5050, 50% sand and 50% silt); (b) the variation of the compression index C_c with the silt content

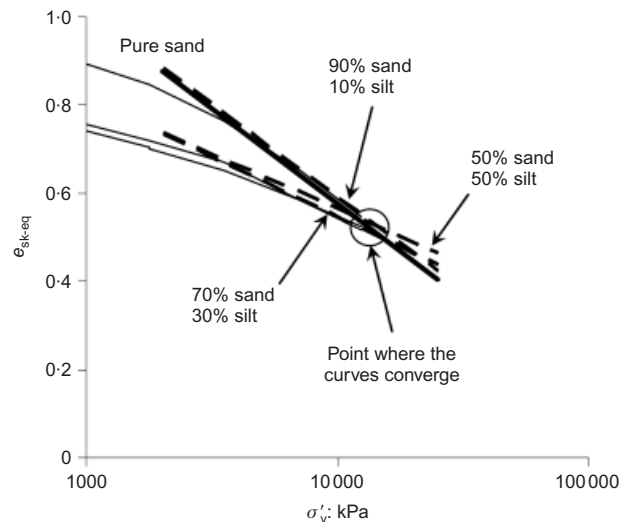


Fig. 5. 1D-NCLs plotted in terms of intergranular void ratio e_{sk-eq} ($b = 0.8$)

assumption would only translate the 1D-NCL rather than change its gradient. Although the critical state data collected from tests on pure silt and on the 5050 material do not reach values of p' high enough to become parallel with the NCLs, the position of the respective 1D-NCLs indicates where the CSLs should be at higher mean effec-

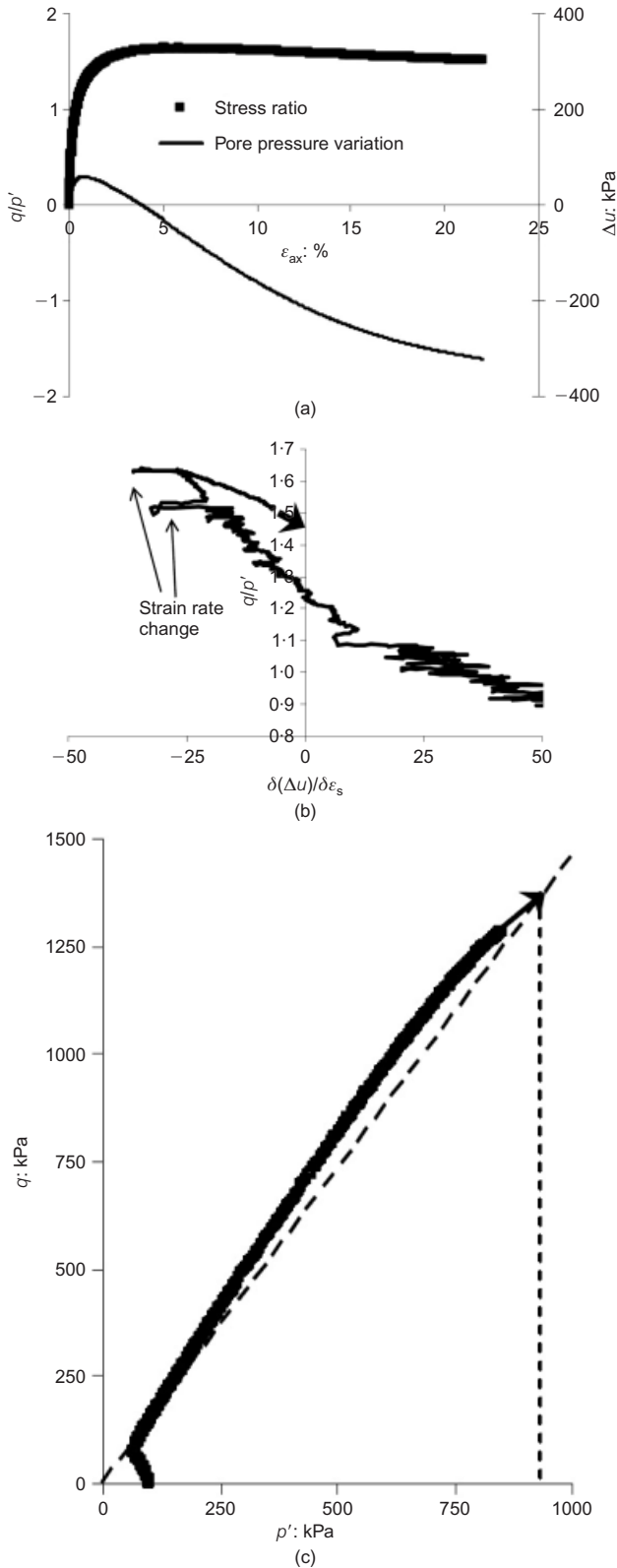


Fig. 6. (a) An example of test (023) that ended before reaching the critical state; (b) estimation of the critical state stress ratio by means of the rate of pore pressure change graph (the two 'disturbances' correspond to a strain rate change during the shearing phase of the test); (c) estimation of p'_{CS} in the q - p' plane by means of the critical state stress ratio

tive stresses. As anticipated from the oedometer tests, no grading shows any significant transitional behaviour and unique CSLs could be found for each, although it can be noticed, again, that the data defining the CSL of the 5050

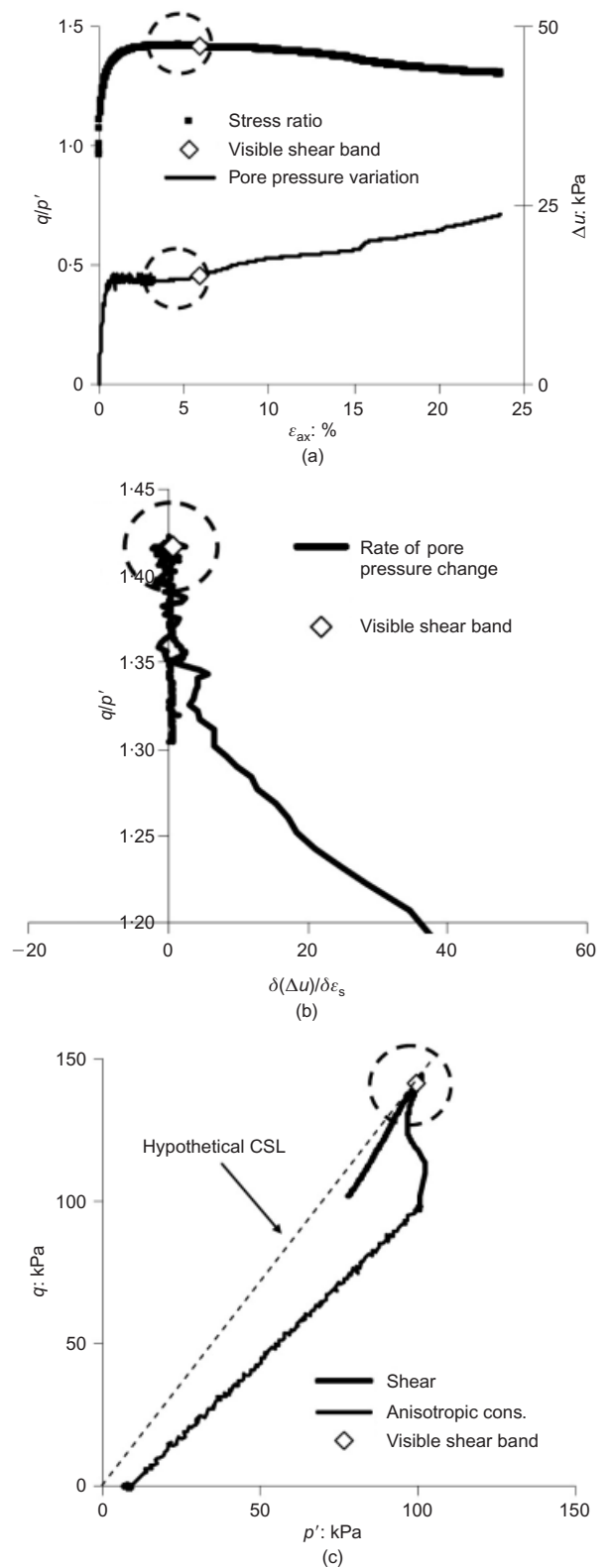


Fig. 7. Triaxial test 005 exhibiting strain localisation: (a) stress ratio and pore pressure variation; (b) rate of pore pressure variation graph; (c) stress path. The circles indicate the zone where the sample almost reached a critical state before showing strong strain localisation

material seem to be slightly more scattered than the others.

From a comparison of the curves (Fig. 10), it can be seen that they all have a very similar initial low gradient, each with a horizontal asymptote, as p' tends towards zero. The

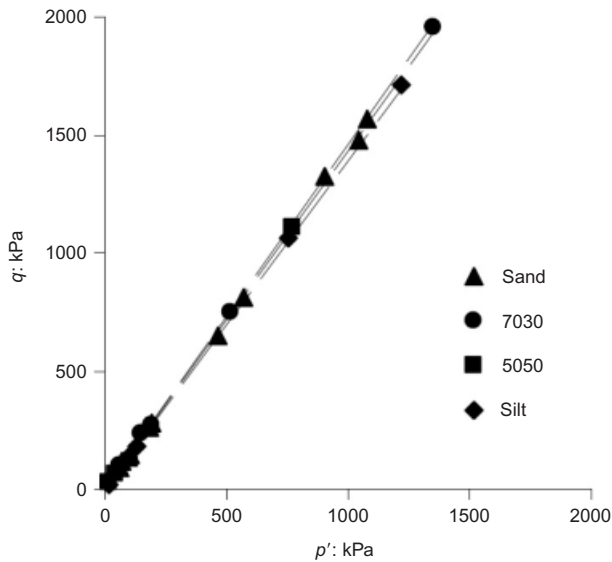


Fig. 8. Critical state points of the soil mixtures tested

increase of silt content moves the CSL downwards, until an inversion of the trend is observed when the silt content is higher than 50%. This behaviour is very similar to that observed by Thevanayagam *et al.* (2002). At higher stresses, the CSLs tend to lie parallel to the 1D-NCLs and are therefore no longer parallel to each other, but have gradients that vary with the fines content as for the 1D-NCLs. It is important, therefore, to note that the gradients of the CSLs at low pressures are unrelated to those at higher stresses. Another important conclusion is that, although the different mixtures have practically the same CSLs in the $q: p'$ plane (Fig. 8), in the $e: \ln(p')$ plane they are significantly different (Fig. 10).

Figure 10 also shows those samples that have been prepared using alternative techniques from the moist tamping, in particular dry pluviation and wet pluviation (indicated with open symbols). The critical state points are in good agreement with those obtained from moist tamped samples, throughout the range of stresses. This independence of the preparation method is in contrast to the results of others, for example Yang *et al.* (2008), who attributed a dependency of

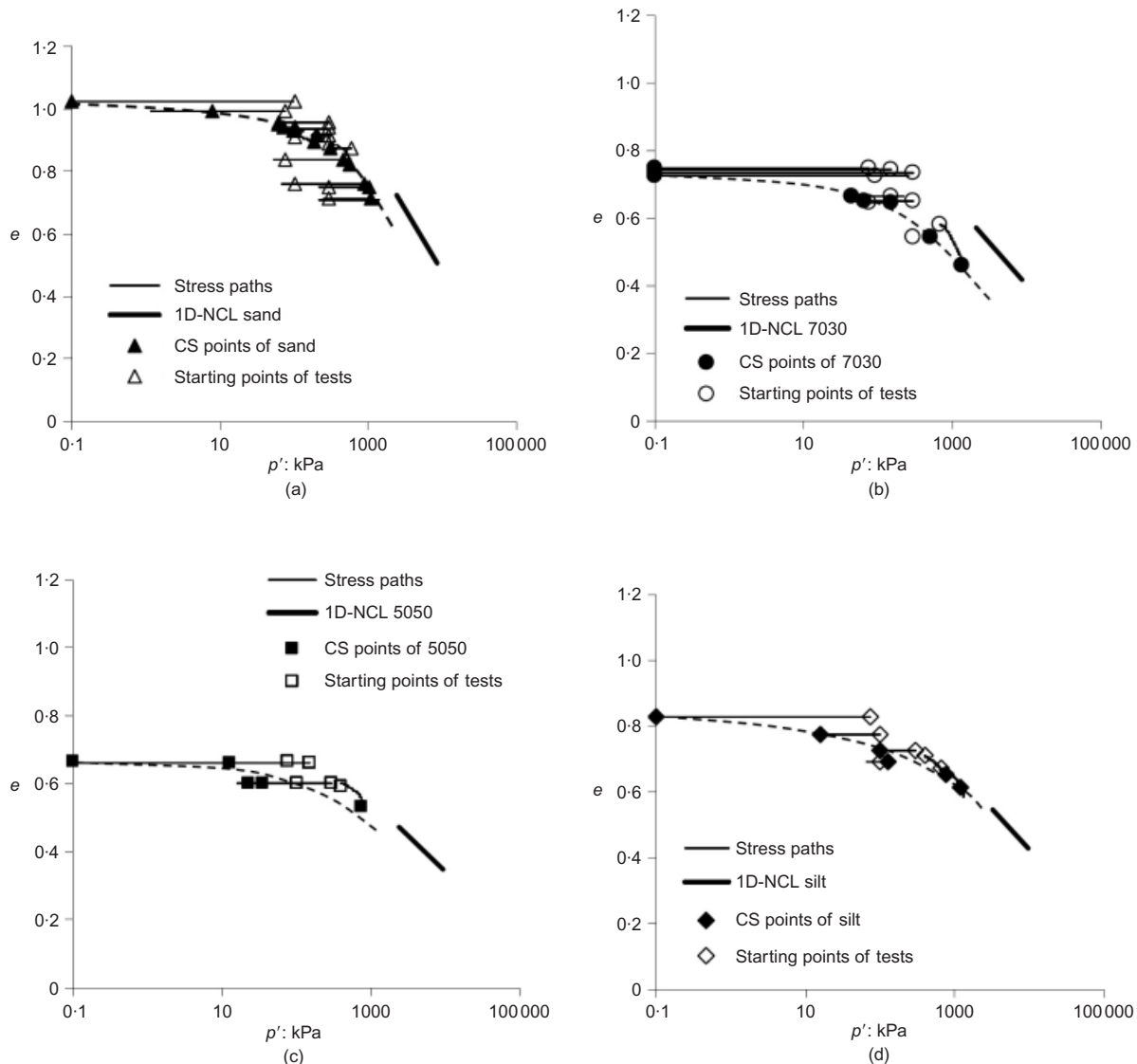


Fig. 9. Initial and critical state points for triaxial tests: (a) on sand; (b) on 7030; (c) on 5050; and (d) on silt. Dotted lines show the most likely shape of the CSLs

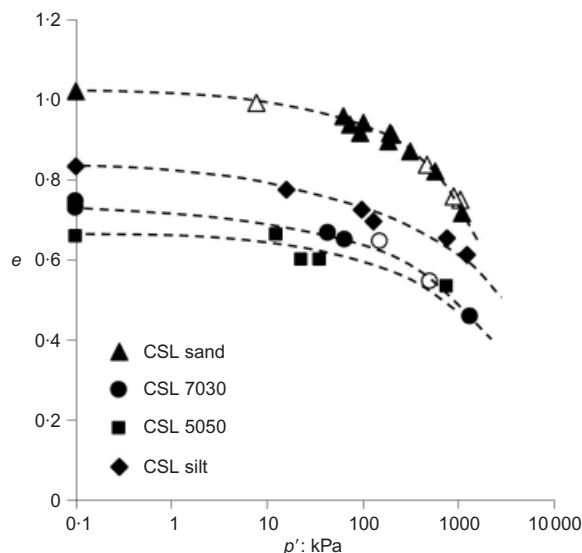


Fig. 10. Critical state data from all the tests. The empty dots correspond to those tests prepared with methods different from moist tamping

the critical state on the sample preparation method to anisotropy arising from grain orientation. However, the grains of the Stava soils are not particularly elongated, as indicated by the QicPic shape analysis data, so that it is possible that any such anisotropy is negligible.

The curvature of the CSL line is often attributed to the onset of the particle breakage (e.g. Konrad, 1998). Particle size distributions were performed on pure sand samples after triaxial shearing, two of them having reached a critical state that is on the flatter part of the CSL, one that has its critical state at the 'knee' and two that reached higher levels of p' . The grading curves are shown in Fig. 11. No appreciable difference can be observed, showing that particle breakage is not a necessary feature of the change in gradient of a CSL.

An interesting feature is observed when the shape of the CSL for each soil mixture is compared with the shape of the oedometer curve corresponding to the loosest sample of the same mixture. As shown in Fig. 12 for the pure sand, the two curves run almost parallel down to low pressures. The same was found to be the case for the 7030 and the 5050 mixtures, but for the pure silt this feature was not seen. Although this needs further investigation, it is of interest in determining the shape of the CSL at low stress levels, in order to identify the risk of liquefaction.

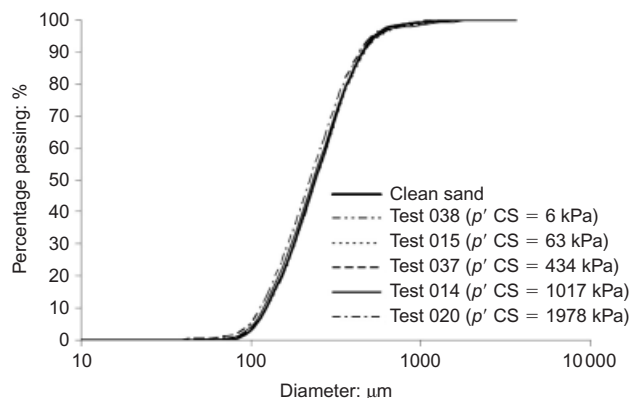


Fig. 11. Comparison of grading curves of sandy samples after triaxial tests

STRAIN SOFTENING AND LIQUEFACTION

For some sands at some combinations of density and stress level, undrained shearing from a state that is initially above the CSL, as in Fig. 1, can be simply strain hardening to a critical state, with compressive behaviour, that is an increase of pore pressure. This seems often to be the case at higher stress levels when samples are sheared from the straight part of the NCL to the straight part of the CSL (e.g. Coop & Lee, 1993). In other cases, for states above the CSL a strain-softening behaviour is observed that leads to uncontrolled failure when the loading is stress controlled. There is a general agreement to call this behaviour 'liquefaction', even if a sample in the same state, if sheared under strain control, does not show any 'unstable' behaviour or liquefaction. More properly, the behaviour should be termed true or complete liquefaction if the specimen visibly appears to liquefy as the pore pressure equals the cell pressure, and p' and therefore q tend to zero.

In the literature, there are only few examples of authors who distinguish clearly between the two definitions of liquefaction (e.g. Yamamuro & Covert, 2001) and it is often hard to understand which definition authors adopt. Fig. 13 shows examples of stress paths for the two different responses obtained during this research. Test 015 suffers a severe strain softening but no liquefaction as it reaches a stable critical state, whereas test 027 was observed to liquefy, and thus its behaviour can be called 'true liquefaction'. This leads to an important consequence, which is that true liquefaction cannot be reached by all samples whose state is above the CSL, but only by those samples whose initial state lies above the horizontal asymptote of the CSL in the $e: \ln(p')$ plane so that, when sheared, p' tends to zero (Fig. 14). Susceptibility to true static liquefaction therefore depends on the initial void ratio of the soil but not on the stress state. Similar behaviour was seen by Yamamuro & Lade (1998).

All the sand-silt mixtures investigated were found to reach 'true liquefaction' providing the samples were sufficiently loose. From the above it can be seen that if a test undergoes true liquefaction at a given void ratio e_{liq} , all specimens at higher void ratios must liquefy too, and this was confirmed by the test data. On Fig. 9, the critical states of all samples that were observed to undergo static liquefaction have been identified with a final data point plotted arbitrarily at 0.1 kPa on the $\ln(p')$ axis.

The minimum void ratios, e_{liq} , at which each soil composition reached liquefaction are shown in Fig. 15; the 5050 soil reaches liquefaction at the smallest void ratio. However, the value of e_{liq} alone is not a sufficient indicator of the susceptibility to liquefaction of a soil, and to see which of the Stava sand-silt mixtures is the most prone to liquefaction, the e_{liq} was compared with the respective e_{max} and e_{min} . While for the sand these were directly measured using the ASTM D4253 and D4254 specifications, this was not possible for the other mixtures because of the presence of silt; in these cases, e_{max} and e_{min} were derived from the initial void ratios of the loosest and of the densest samples that it was found possible to create in the oedometer tests. These values of e_{max} and e_{min} are therefore only a qualitative estimate. Nonetheless, they are likely to give a good indication of the true values, since the e_{max} and e_{min} corresponding to the loosest and densest samples of the sand were very similar to those from the ASTM tests.

The indicative e_{max} and e_{min} and the e_{liq} of the different soils in Fig. 15 show that the sandy soil needs to be nearly at its loosest state to reach liquefaction. The other mixtures only need to be in a state closer to a medium density within the range, although there is no clear trend since the 5050 material seems to be again less susceptible to liquefaction

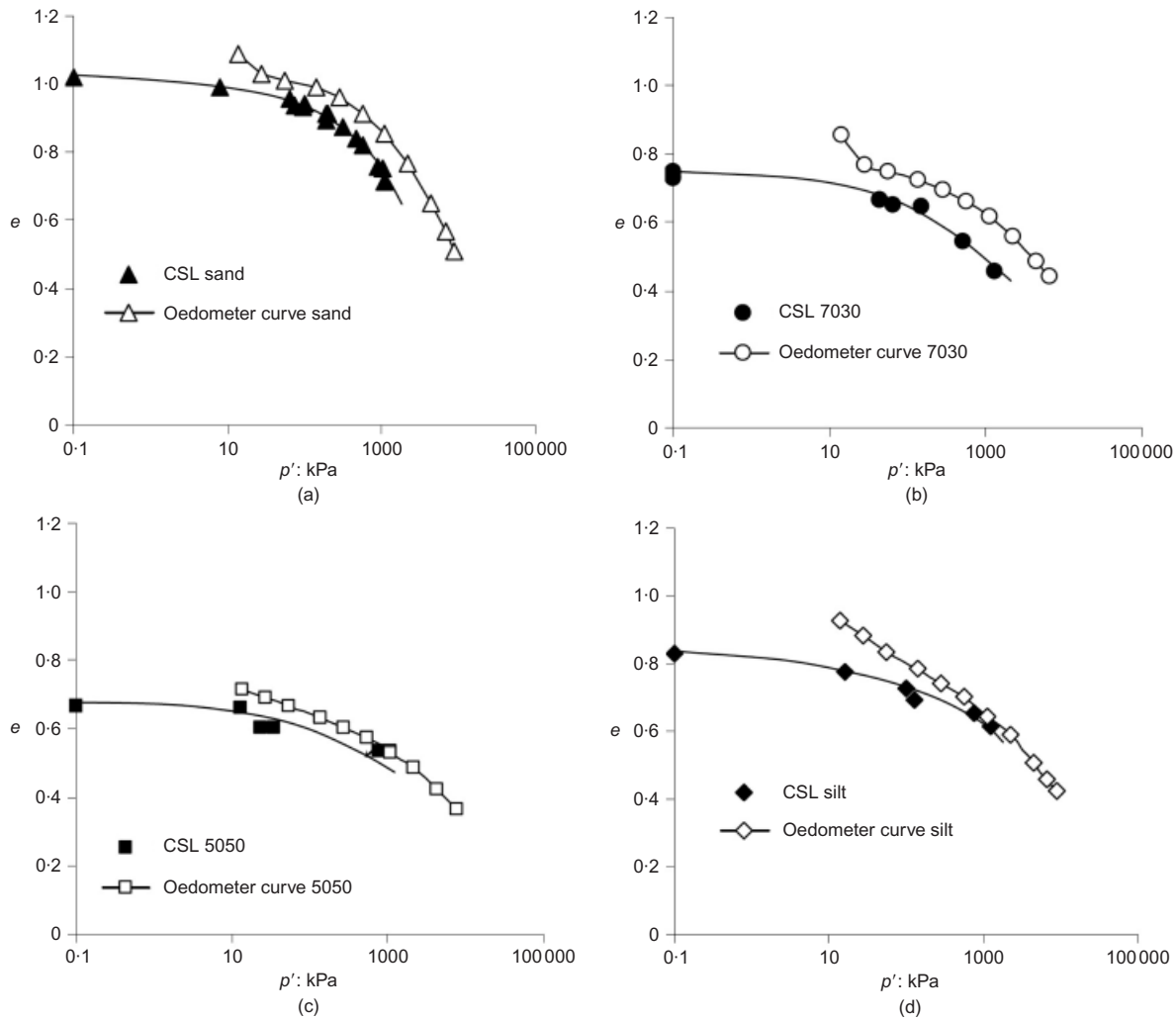


Fig. 12. Comparison between the CSLs and the oedometer curves obtained from samples with maximum e_0 : (a) sandy material; (b) 7030 mixture; (c) 5050 mixture; (d) silt

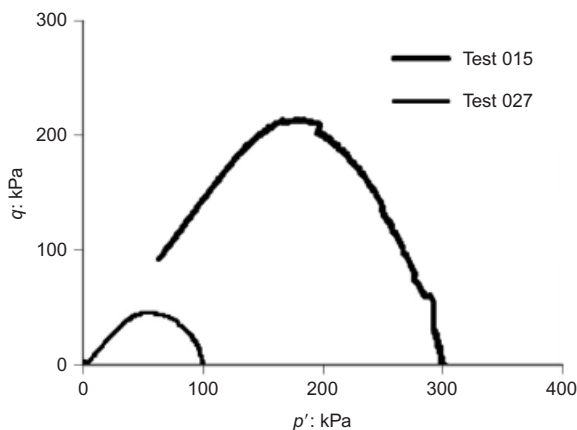


Fig. 13. Comparison between two tests on sand: test 015 shows strain softening, test 027 is subject to 'true liquefaction'

than the 7030 or 100% silt. In general, the difference between the maximum void ratio and that required for liquefaction is fairly similar for each grading, but the range of achievable void ratios varies with much wider ranges for the less susceptible materials, that is pure sand and 5050. It is the changing range that has the biggest influence on the relative density required for liquefaction. A comparison of Fig. 15 with Fig. 9 and Fig. 4 shows that the locations of

the CSLs or NCLs in the $e: \ln(p')$ plane on their own are not a good indicator of liquefaction susceptibility and that it is the relative locations of achievable states and these lines that is important. These results are in agreement with those obtained by Lade & Yamamuro (1997), who used sand mixed with non-plastic fines, and showed that the presence of non-plastic fines increases the susceptibility to liquefaction of a sand. If the $e_{\max}$ and $e_{\min}$ that have been considered are correct, it should be also noted that the CSL is not asymptotic to the value of $e_{\max}$ in the $e: \ln(p')$ plane as is often assumed (e.g. Gudehus, 1996), but to a lower void ratio, as was also found by Yamamuro & Covert (2001). It would not be appropriate to make detailed comments in this paper about the causes of the failure of the Stava dams, but from Fig. 15, it can be seen that the materials with the higher silt contents (particularly 7030 and pure silt) are more susceptible to liquefaction than the clean sand. These finer materials would be more typically found in the beach and lagoon; however, the stability of the structure would also depend on whether the materials were saturated and also on their in situ density, which have not been addressed in this paper.

It therefore appears that an embankment made of clean sand would be more resistant to liquefaction than one constructed of the sand mixed even with a small percentage of fines. It is also clear that adequate compaction of the sandy material during the construction of the embankment is of paramount importance. As for many other similar tailings

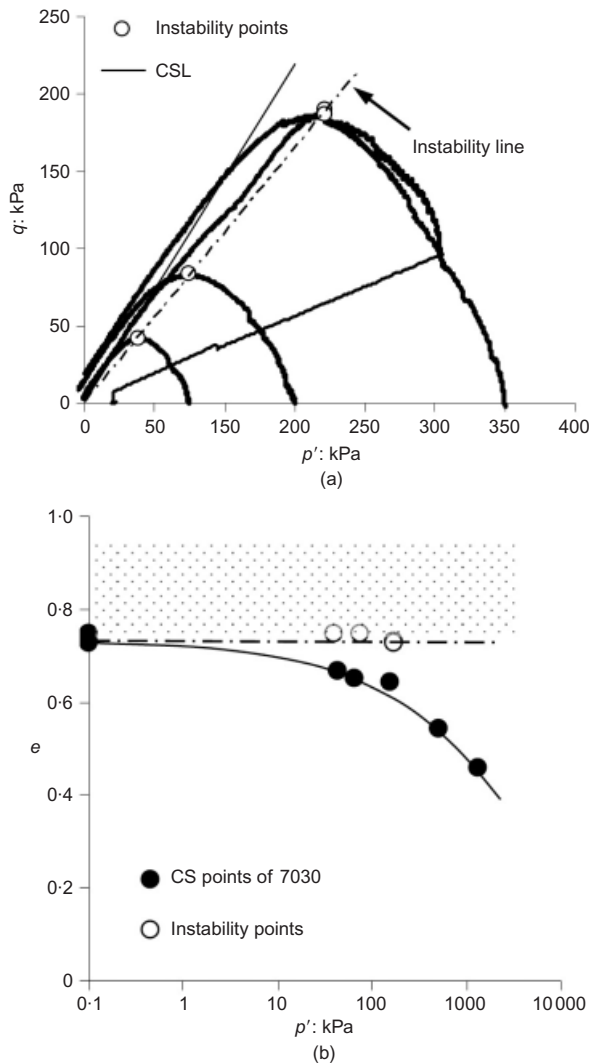


Fig. 14. (a) Stress paths and locus of the $q_{\max}$ for tests 017, 029 and 032 on 7030; (b) instability points in the compression plane for the 7030 mixture (the shaded zone is where the instability points can be found)

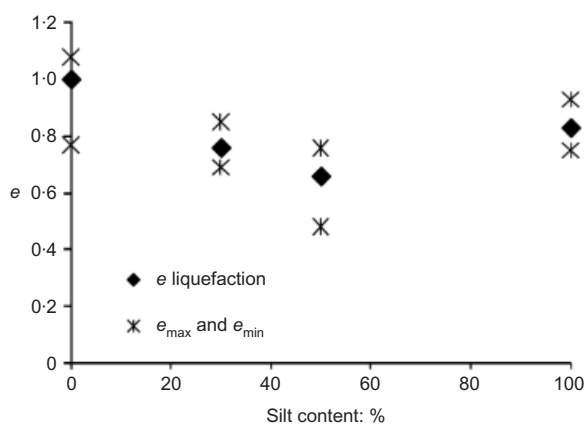


Fig. 15. Void ratio of the samples that reached true liquefaction at different silt contents and respective maximum and minimum void ratios ($e_{\max}$ and $e_{\min}$)

dams, the embankment of the Stava tailings dam was constituted of sand separated from the silt by means of a hydro-cyclone, so the resulting sand contained between 10% and 40% silt. It was ejected in a moist state and probably only lightly compacted and so the relative density might

have been quite low. The silty material that constituted the basin was mostly in a saturated state, and so being also placed in a loose state would have been susceptible to liquefaction too.

The position of the instability line was investigated and compared to the CSL location in Fig. 14. The term 'instability line' will be taken here to refer to the locus of $q_{\max}$ points only of those samples that reached true liquefaction. Four tests on the 7030 mixture (tests 017, 029, 032 and 036) were considered, three of them isotropically consolidated to p'_c values ranging from 57 to 300 kPa, and one anisotropically consolidated. The void ratios after compression, e_c , of the four samples varied between 0.748 and 0.727. Fig. 14(a) shows that in the $q: p'$ plane the instability line is unique, regardless of e_c and p'_c . In the $e: \ln(p')$ plane, the instability points must all lie above the horizontal asymptote of the CSL, that is in the shaded zone in Fig. 14(b), since only samples with void ratios greater than e_{liq} will undergo static liquefaction. Clearly, if the void ratio of two samples is the same, then their instability points will lie on the same horizontal line.

The positions of the $q_{\max}$ points for those tests that show strain softening but no liquefaction was then investigated. Fig. 16(a) shows the stress paths of the five tests on clean sand, which showed this behaviour and differ in their consolidation pressures and void ratios. Only one of them reached true liquefaction (test 27 at $p'_c = 100$ kPa), for

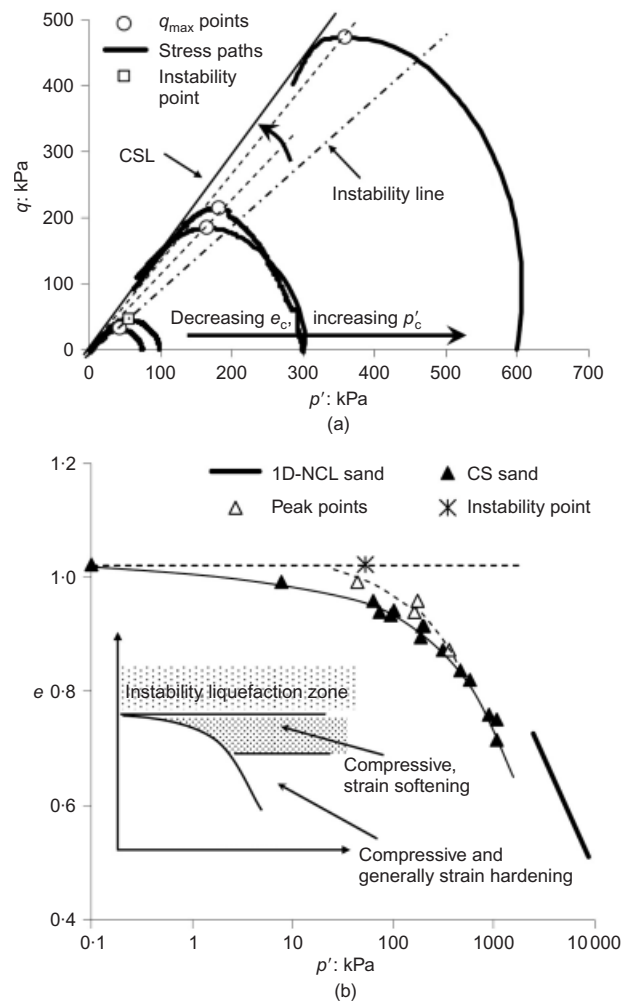


Fig. 16. (a) Stress paths and locus of $q_{\max}$ from tests on clean sand that showed strain softening; (b) a suggested scheme of subdivision of the area above the CSL into three different regions connected to different behaviour

which the $q_{\max}$ point is then an instability point and lies on the instability line. The $q_{\max}$ points tend to move towards the CSL in the $q: p'$ plane and the reduction in q after peak tends to decrease as the void ratio decreases and as the mean effective stress increases. The instability line and the CSL seem to limit the zone where the $q_{\max}$ can be found in the $q: p'$ plane. Similar behaviour has been observed by Chu & Leong (2002) and Yang (2002), among others, although most authors refer to such $q_{\max}$ points as instability points regardless of whether the sample goes to true liquefaction or not.

In the $e: \ln(p')$ plane (Fig. 16(b)), the $q_{\max}$ points can be found only in a certain range of e above the CSL and below the horizontal asymptote of the CSL that bounds the instability zone. No $q_{\max}$ points are found below a certain value of e that seems to correspond to the point where the CSL tends to a straight line, and at higher pressures, no peaks can be found for undrained tests on the wet side of the CSL, as Coop & Lee (1993) had seen at high stress levels. There seems to be a locus of $q_{\max}$ points that is indicated with a broken line from the horizontal asymptote of the CSL towards the straight high pressure part of the CSL.

CONCLUSIONS

The oedometer tests showed that all the sand and silt mixtures studied reached unique 1D-NCLs, and that none of them has transitional behaviour despite having gradings similar to those of other transitional soils. However, the oedometer compression curves obtained for the soil composed of 50% sand–50% silt were much more scattered than the others, which was recognised as the possible onset of transitional behaviour. The location of the 1D-NCL was found to move downwards and to reduce its inclination in the $e: \log(\sigma'_v)$ plane as the silt content was increased, the trend being reversed when the silt content was higher than 50–70%. The average void ratio that was achieved preparing the samples also tended to decrease with the increase of silt content, but increase again for silt contents over 50%.

The use of an equivalent intergranular void ratio e_{sk-eq} was found to give some convergence for the oedometer curves with a silt content up to 50% if the value of the parameter b was equal to 0.8. However, closer inspection showed that any convergence was only apparent and the 1D-NCLs defined in terms of e_{sk-eq} actually cross each other. This might be expected, since b has no real physical meaning but simply gives a qualitative idea of to what extent the fines are involved in sustaining the stresses.

The CSLs of each soil tested were found through triaxial tests. The critical state angle of shearing resistance was found to be very similar for all the soils investigated, probably due to their common origin and thus similar mineralogy and particle shapes for all gradings. In the $e: \ln(p')$ plane the CSLs of all the mixtures showed some degree of curvature, and the gradient of the CSLs at lower pressures was found to be independent from that at higher pressures. At low–medium pressures, the CSLs are fairly parallel, fairly flat and only translate downwards and upwards as the silt content is changed, whereas at higher pressures, the slopes of the straight part of the CSLs change significantly, as is true also for the straight part of the NCLs. It also seems that, for materials composed of a medium–high content of sand, the CSL of a soil runs parallel to the oedometer curve obtained from the loosest sample of that soil. Grain size distributions carried out after triaxial tests on the pure sand showed that no appreciable crushing took place, even at high pressures, which means that the curvature of the CSL cannot be due only to the onset of the

particle breakage as sometimes supposed in the literature. No differences in the critical states were observed for samples prepared with different methods, perhaps because the shape of the particles is quite regular.

In studying the susceptibility to liquefaction of the soils, only those samples that reached a state in which p' and q tended to zero were considered as having undergone liquefaction and it was found that this occurred only for samples whose void ratio was higher than a certain e_{liq} , regardless of the stress state, where the e_{liq} corresponds to the horizontal asymptote of the CSL in the $e: \ln(p')$ plane. To see which mixture of sand and silt was most susceptible to liquefaction, the values of e_{liq} were compared with the maximum and minimum void ratios. It appears that a silt content of about 30% is the most susceptible, as the relative density at which it reaches liquefaction is higher than those required for other mixtures. This has the consequence that a dam embankment made of sand mixed with a relatively small quantity of fines (as it usually is when the sand is separated from the fines using a hydro-cyclone) is significantly more prone to liquefaction, even if the void ratio is lower than for the equivalent clean sand.

The position of the instability line was also investigated. The results showed that if instability is defined as the $q_{\max}$ of samples that subsequently liquefy, it is independent of the stress level and the void ratio. Instead, the position of the $q_{\max}$ points of those samples that showed strain softening but not liquefaction is dependent on void ratio and stress level. The $q_{\max}$ points seem to be closer to the CSL in the $q: p'$ plane for higher consolidation stresses and lower void ratios. In the $e: \ln(p')$ plane three zones of behaviour can be envisaged, as shown in the schematic diagram in Fig. 16(b). Samples with initial states above e_{liq} , which corresponds to the horizontal asymptote of the CSL, will have instability points followed by true liquefaction. Samples in the next zone down will reach stable critical states on the curved part of the CSL when sheared undrained, and will be compressive and strain softening after a point of $q_{\max}$. At the highest stresses, the samples will reach stable critical states on the straight part of the CSL and will be compressive and generally strain hardening.

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NOTATION

1D-NCL	one-dimensional normal compression line
C_c	compression index
CSL	critical state line
e	void ratio
e_{sk-eq}	equivalent intergranular void ratio
G_s	specific gravity
K_0	coefficient of earth pressure at rest
PI	plasticity index
p'	mean normal effective stress $= (\sigma'_1 + 2\sigma'_3)/3$
p'_{cs}, q_{cs}, e_{cs}	mean effective stress, deviatoric stress and void ratio at the critical state
p'_c, q_c, e_c	mean effective stress, deviatoric stress and void ratio at the end of compression

q	deviator stress = $(\sigma'_1 - \sigma'_3)$
q/p'	stress ratio
$(q/p')_{CS}$	stress ratio at the critical state
TFC	transitional fines content
v	specific volume = $(1 + e)$
w_L	liquid limit
w_P	plastic limit
$\Delta\varepsilon_v/\delta\varepsilon_s$	dilatancy
$\delta(\Delta u)/\delta\varepsilon_s$	dilatancy in terms of pore pressure change
φ_{CS}	angle of shearing resistance at the critical state

REFERENCES

- ASTM (2002a). D 4253-00: Standard test methods for maximum index density and unit weight of soils using a vibratory table. West Conshohocken, PA: ASTM International.
- ASTM (2002b). D 4254-00: Standard test methods for minimum index density and unit weight of soils and calculation of relative density. West Conshohocken, PA: ASTM International.
- Been, K. & Jefferies, M. G. (1985). A state parameter for sands. *Géotechnique* **35**, No. 2, 99–112, doi: 10.1680/geot.1985.35.2.99.
- Been, K., Jefferies, M. G. & Hachey, J. (1991). The critical state of sands. *Géotechnique*, **41**, No. 3, 365–381, doi: 10.1680/geot.1991.41.3.365.
- Blight, G. E. & Fourie, A. B. (2003). A review of catastrophic flow failures of deposits of mine waste and municipal refuse. *Proceedings of the international conference FSM2003*, Naples, 1–17.
- Chandler, R. J. & Tosatti, G. (1995). The Stava tailings dams failure, Italy, July 1985. *Proc. Instn Civ. Engrs, Geotech. Engng* **113**, No. 2, 67–79.
- Chu, J. & Leong, W. K. (2002). Effect of fines on instability behaviour of loose sands. *Géotechnique* **52**, No. 10, 751–755, doi: 10.1680/geot.2002.52.10.751.
- Chu, J., Leong, W. K. & Loke, W. L. (2003). Discussion of 'Defining an appropriate steady state line for Marriespruit gold tailings'. *Can. Geotech. J.* **40**, No. 2, 484–486.
- Colombo, P. & Colleselli, F. (2003). The collapse of the Stava tailings basins – analysis of the event and observations. *Int. Workshop on Occurrence and Mechanisms of Flows in Natural Slopes and Earthfills, Sorrento, Italy* (ed. L. Picarelli), pp. 87–96. Bologna: Patron.
- Coop, M. R. & Atkinson, J. H. (1993). The mechanics of cemented carbonate sands. *Géotechnique* **43**, No. 1, 53–67, doi: 10.1680/geot.1993.43.1.53.
- Coop, M. R. & Lee, I. K. (1993). The behaviour of granular soils at elevated stresses. In *Predictive soil mechanics* (eds G. T. Houlsby and A. N. Schofield), Proceedings of the C. P. Wroth Memorial Symposium, 1992, pp. 186–198. London: Thomas Telford.
- Ferreira, P. M. V. & Bica, A. V. D. (2006). Problems in identifying the effects of structure and critical state in a soil with a transitional behaviour. *Géotechnique* **56**, No. 7, 445–454, doi: 10.1680/geot.2006.56.7.445.
- Fukushima, S. & Tatsuoka, F. (1984). Strength and deformation characteristics of saturated sand at extremely low pressure. *Soils Found.* **24**, No. 4, 30–48.
- Gudehus, G. (1996). A comprehensive constitutive equation for granular materials. *Soils Found.* **36**, No. 1, 1–12.
- Jefferies, M. G. & Been, K. (2006). *Soil liquefaction – a critical state approach*. Taylor and Francis.
- Konrad, J. M. (1998). Sand state from cone penetrometer tests: a framework considering grain crushing stress. *Géotechnique* **48**, No. 2, 201–215, doi: 10.1680/geot.1998.48.2.201.
- Kramer, S. L. (1996). *Geotechnical earthquake engineering*. Upper Saddle River, NJ: Prentice Hall.
- Lade, P. V. & Yamamuro, J. A. (1997). Effects of nonplastic fines on static liquefaction of sands. *Can. Geotech. J.* **34**, No. 6, 918–928.
- Mitchell, J. K. (1976). *Fundamentals of soil behaviour*. New York: John Wiley.
- Ni, Q., Tan, T. S., Dasari, G. R. & Hight, D. W. (2004). Contribution of fines to the compressive strength of mixed soils. *Géotechnique* **54**, No. 9, 561–569, doi: 10.1680/geot.2004.54.9.561.
- Nocilla, A., Coop, M. R. & Colleselli, F. (2006). The mechanics of an Italian silt: an example of 'transitional' behaviour. *Géotechnique* **56**, No. 4, 261–271, doi: 10.1680/geot.2006.56.4.261.
- Pitman, T. D., Robertson, P. K. & Sego, D. C. (1994). Influence of fines on the collapse of loose sands. *Can. Geotech. J.* **31**, 728–739.
- Polito, C. P. (1999). *The effects of non-plastic and plastic fines on the liquefaction of sandy soils*. PhD thesis, Virginia Polytechnic Institute, USA.
- Polito, C. P. & Martin, J. R. II (2001). Effects of nonplastic fines on the liquefaction resistance of sands. *J. Geotech. Geoenviron. Engng* **127**, No. 5, 408–415.
- Rahman, M. M. & Lo, S. R. (2008). The prediction of equivalent granular steady state line of loose sand with fines. *Geomech. Geoenng: Int. J.* **3**, No. 3, 179–190.
- Sladen, J. A., D'Hollander, R. D. & Krahn, J. (1985). The liquefaction of sands: a collapse surface approach. *Can. Geotech. J.* **22**, No. 4, 564–578.
- Thevanayagam, S., Shenthan, T., Mohan, S. & Liang, J. (2002). Undrained fragility of clean sands, silty sands and sandy silts. *J. Geotech. Geoenviron. Engng ASCE* **128**, No. 10, 849–859.
- Vaid, Y. P. & Sivathayalan, S. (2007). Fundamental factors affecting liquefaction susceptibility of sands. *Can. Geotech. J.* **37**, No. 3, 592–606.
- Verdugo, R. & Ishihara, K. (1996). The steady state of sandy soils. *Soils Found.* **36**, No. 2, 81–91.
- Yamamuro, J. A. & Covert, K. M. (2001). Monotonic and cyclic liquefaction of very loose sands with high silt content. *J. Geotech. Geoenviron. Engng* **127**, No. 4, 314–324.
- Yamamuro, J. A. & Lade, P. V. (1998). Steady-state concepts and static liquefaction of silty sands. *J. Geotech. Geoenviron. Engng* **124**, No. 9, 868–877.
- Yang, J. (2002). Non-uniqueness of flow liquefaction line for loose sand. *Géotechnique* **52**, No. 10, 757–760, doi: 10.1680/geot.2002.52.10.757.
- Yang, S. (2004). *Characterization of the properties of sand-silt mixtures*. PhD thesis, Norwegian University of Science and Technology, Norway.
- Yang, Z. X., Li, X. S. & Yang, J. (2008). Quantifying and modelling fabric anisotropy of granular soils. *Géotechnique* **58**, No. 4, 237–248, doi: 10.1680/geot.2008.58.4.237.